SYSTEM & MAINTENANCE MANUAL

FOR

SECTIONAL DOOR

CLIENT: RASS PROJECTS CONSULTANT PVT LTD

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DESIGNED, DEVELOPED & MARKETED

BY

NIHVA Technologies Pvt. Ltd.

G-BLOCK, 1ST FLOOR, THERMAX CHOWK

MIDC, ABOVE RENAULT SHOWROOM

CHINCHWAD, PUNE-411019

TEL.NO: - +912066304033

EMAIL: -sales@nihva.com

www.nihva.com

ABTRACT:

We are pleased to forward this manual to you to provide guidelines on trouble free operation of the equipment, procedures for periodic and preventive maintenance, and troubleshooting. This manual consists of the description of equipment along with technical specifications, and operation procedures. For maintenance of 'Bought-out' equipment, please refer manufacturers' manuals. We insist you to maintain necessary record (logbook) for ensuring proper preventive maintenance, to achieve trouble free production. We are sure; you will find this manual very useful and interesting. However, your suggestions and doubts if any are always welcome!

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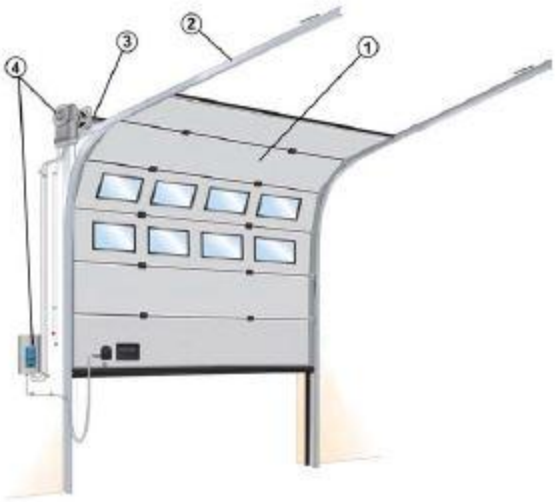
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1. Introduction

The Overhead sectional door is designed to enable easy access to and from buildings.
The door leaf is made of insulated panels, or a combination of insulated panels and acrylic windows. The door leaf moves along tracks installed at each side of the door leaf. It can be operated manually, or by a chain hoist or an electrical door control system.



The door has 4 primary parts:

1) Door leaf
2) Track set
3) Balancing system
4) Operating system/chainhoist (optional)

1.1 Door leaf

The door leafs are an assembly of horizontal sections connected together via hinges. The outer hinges of each section have rollers that engage in the tracks to enable easy opening and closing of the door.
There are two types of door leaf; one is made of sandwich panels, the other is made of framed window panels. Glazed sections can be installed in the sandwich panels if required.
Seals installed on the sides, top and bottom of the door leafs improve the insulation of the working environment. The bottom seal also acts as a shock absorber when the door closes.
A pull-down rope and a handle are installed on the door leaf to enable easy opening and closing of the door. A door lock is also installed on the door for extra security of the working environment.
A pass-door can be installed in the door leaf to permit pedestrian access without opening the door.

1.2 Track set

The track set supports the door leaf. The rollers on the door leaf move along the track set when the door is opened and closed. Different track sets can be installed dependent on the structure of the building and the space available.

1.3 Balancing system

The balancing system balances the weight of the door to enable easy opening and closing of the door. It also enables the door to stay in a partly open position if necessary.

The system is installed on the top of the track set and consists of torsions springs installed on a shaft that has a cable drum at each end. Cables on the cable drums are connected to the bottom corners of the door leaf.

When the door is opened, the torsion springs turn the shaft with a force equal to the weight of the door and this pulls the door up as far as necessary.

A spring break device is installed for each torsion spring on the shaft. If a torsion spring breaks the spring break device prevents the door closing.

An optional cable break device can also be installed to prevent the door closing if a cable breaks.

1.4 Operating system

1.4.1 Chain hoist

For heavier doors, chain hoist operation is easier than manual operation.

There are three types of chain hoist:

- D-hoist: Non-geared chain transmission directly connected to the shaft.
- T-hoist: Geared (ratio 1:4) chain transmission indirectly connected to the shaft.
- U-hoist: Geared (ratio 1:3) chain transmission connected to another chain between the chain hoist and the shaft.

An example of a U-hoist is shown below:



1.4.2 CDM9

The CDM9 permits opening and closing of the door via a push-button control unit. It is also possible to enable use of Access and Automation functions.

The primary parts of the CDM9 are the operator (electric motor) that is installed on the balancing shaft, and the door control system.

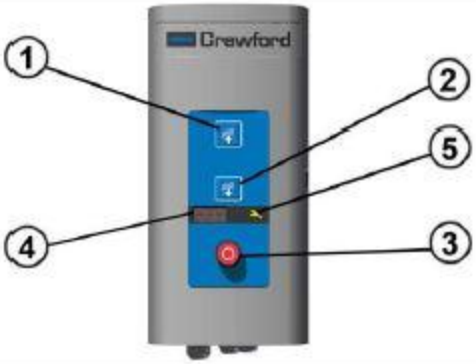




There are three models of the door control system:

- 920 Door control system: Has hold-to-run or impulse open, hold-to-run close, and stop functions. Also a slot for an external control box.
- 930 Door control system: Impulse open and close function (930 has always a safety edge installed on the door).
- 950 Door control system: Has the same functions as the 930 plus a digital display for troubleshooting and maintenance purposes. It can also be connected to Access and Automation functions.





1) Up button
2) Down button
3) (Emergency) stop button
4) Digital display for troubleshooting and maintenance (only 950)
5) Service interval indicator (only 950)



The number on the display shows numbers of door openings x1000. Yellow led in the lid indicates that the maintenance interval has been achieved. Factory setting is 20 000 cycles/365 days.

1.4.3 Technical specifications

Classification	IP 55 (Excluding adapter which is IP44)
Supply voltage Operator	230V, 1~
Supply voltage Hydraulic	230V 3~, 400V 3N~
Temperature range	-20°C to +55°C
Operating factor Operator	ED = 30%
Operating factor Hydraulic	ED = 10%
Atmospheric humidity	0-80% relative, not condensing

2. Safety

2.1 General safety statements

The door has been designed to meet all operational and safety requirements in the European Directives and the standards issued by the European Standardization Committee, CEN.

2.2 Safety instructions



Important safety instructions.

- It is important for the safety of persons to follow these instructions. Save these instructions.
- Only authorized persons are permitted to operate the door.
- Make sure there are no persons or equipment in the working area of the door before you operate the door.
- This door is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the door by a person responsible for their safety.
- Children should be supervised to ensure that they do not play with the door or controls. Keep remote controls away from children.
- Frequently examine the installation for imbalance and signs of wear or damage to cables, springs and mounting. Do not use the door if repair or adjustment is necessary. Report defects immediately.
- Electrically disconnect and immobilize the door before you do maintenance on the door.
- Do not use the door leaf or track set to support a ladder when you do maintenance on a door. Always use ladders as specified in local health and safety instructions.
- Do not operate the door after the date of the next scheduled maintenance. The date of the next scheduled maintenance is shown in the logbook.
- If one of the safety devices has been activated, take the door out of operation and contact the service centre immediately.
- If the supply cord is damaged, it must be replaced by a special cord or assembly available from the manufacturer or its service centre.
- Do not remove or immobilize safety equipment installed on or near the door.
- Activation of the manual release may cause uncontrolled movement of the driven part due to mechanical failures or an out-of-balance condition.
- Do not modify, adjust or disassemble any part of the overhead sectional door, including the door cables, balancing system and cable attachments. Unauthorized modification can cause danger to people and affect the function and safety of the door.
- Do not attempt to balance the door yourself. Forces in the system can cause serious personal injury.
- Always obey local health and safety regulations when you operate or do maintenance on the door.

2.3 Safety Symbols Used in this Manual

The following safety symbol is used in this manual:



Indicates a general warning

2.4 Safety Information

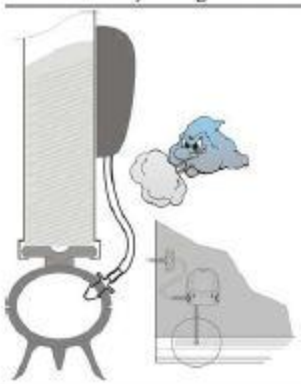
2.4.1 Pneumatic edge

The bottom sealing is connected to a pneumatic switch via a hose. If the sealing is compressed while the door is on its way downward, the circuit in the control unit reacts by reversing the door (during the last 5 centimeters before closed position, when the torque monitor works as safety, it will only stop).The safety edge is supervised. This means that the safety edge is tested each time the door is closed. If the door leaf has passed the lower limit without detection of a pulse from the pneumatic switch, the control unit goes into hold -to-run function. The door can now only be closed by pressing the close-button continuously.



To avoid injury during operation of the excitation unit, monitor the door's closing movement until the door is closed.

The safety edge can be reset automatically if the safety edge is working and issues a signal at the lowest position (< 50 mm from the floor) during hold-to-run operation downwards.



2.5 Safety Procedures

2.5.1 Emergency Stop Procedure (CDM9)

If necessary, pull the rope with the green ball on the end to disengage the operator clutch.



1) When the area is safe:
   1) Connect the electrical supply to the door.
   2) If necessary, pull the rope with the red ball on the end to engage the operator clutch.
   3) Move the door leaf approximately 10 cm up and down to make sure the operator clutch gets into gear.
   4) Reset the emergency stop button (if applicable) and enable normal operation of the door.

2.5.2 Open/Close Door at Electrical Supply Failure (CDM9)

1) Disconnect the electrical supply from the door.



Make sure no persons or equipment are below the door before you disengage the operator clutch. If the door is not correctly balanced it may move down rapidly.

2) Pull the rope with the green ball to disengage the operator clutch.
3) Use the rope and handle to open or close the door.
4) When the electrical supply is available:
   1) Connect the electrical supply to the dock leveller.
   2) Pull the rope with the red ball to engage the operator clutch.
   3) Use the CDM9 to open or close the door.

3. Operating Instructions

3.1 Daily procedures

3.1.1 Daily Start Procedure

1) Examine the door for damage.
2) If damage is found:
   1) Put suitable warning signs around the door.
   2) Contact the applicable person.
3) Electrical operation only:
   1) Switch on the electrical supply to the door.
   2) Reset the emergency stop button (if applicable) and enable normal operation of the door.
   3) If the power was switched off, normal operation is performed after the door has reached its upper position.

3.1.2 Daily Stop Procedure

1) Examine the door for damage.
2) If damage is found:
   1) Put suitable warning signs around the door.
   2) Contact the applicable person.

3.1.3 Open and close the door (920 door control system)

3.1.3.1 Open the door

1) Make sure the Daily Start Procedure has been done.



Make sure there are no persons or equipment in the working area of the door before you operate the door.

2) Make sure the pass-door is closed correctly.
3) Release the door lock.
4) Press and release the  button and make sure the door moves to the fully open position.

3.1.3.2 Close the door



Make sure there are no persons or equipment in the working area of the door before you operate the door.

1) Press and hold the  button until the door is closed.
2) Release the  button.
3) Fasten the door lock.

3.1.4 Open and close the door (930 and 950 door control system)

3.1.4.1 Open the door

1) Make sure the Daily Start Procedure has been done.


Make sure there are no persons or equipment in the working area of the door before you operate the door.


Do not open or close the door if the maintenance indicator is on (950 only). Make sure the pass-door is closed correctly.

2) Release the door lock.
3) Press and release the  button and make sure the door moves to the fully open position.

3.1.4.2 Close the door


Make sure there are no persons or equipment in the working area of the door before you operate the door.


Do not open or close the door if the maintenance indicator is on (950 only).

1) Press and release the  button and make sure the door moves to the fully closed position.
2) Fasten the door lock.
3) Check if the additional safety features such as photocell, magnetic loop, radar, etc are working

3.1.5 Open and close the door (chain hoist)

3.1.5.1 Open the door

1) Make sure the Daily Start Procedure has been done.


Make sure there are no persons or equipment in the working area of the door before you operate the door.

2) Make sure the pass-door is closed correctly.
3) Release the door lock.
4) Carefully pull down on the rear chain until the door is in the open position.

3.1.5.2 Close the door


Make sure there are no persons or equipment in the working area of the door before you operate the door.

1) Carefully pull down on the front chain until the door is in the closed position.
2) Fasten the door lock.

4. Maintenance

In this chapter you find the maintenance tasks that you as a user must perform. With the exception of inspections and maintenance tasks explicitly described in this manual as being performable by the user, all other maintenance tasks, as described in the Logbook are rated advanced and can only be carried out safely by a competent person. Contact your local Service Centre.

4.1 Preventive maintenance schedule

Frequency	Part	Tasks
Daily	Door complete	Do the Daily Start Procedure.
		Do the Daily Stop Procedure.
Monthly	Door complete	If necessary, clean the door leaf
		Do the monthly examination of the door
Every two months	Door complete	If necessary, clean the door leaf
		Do the monthly examination of the door
	Balancing system	Test the balancing system
Every six months	Door complete	Clean the door leaf

4.2 Preventive maintenance procedures

4.2.1 Clean the door leaf



Do not use the door leaf or the track set to support a ladder when you do maintenance on a door. Always use ladders as specified in local health and safety instructions.

1) Do the Daily Stop Procedure.
2) Use a soft clean brush and mild detergent to clean the inside and outside of the door leaf.
3) Examine the painted surfaces of the door leaf for damage to the paint.
4) If damage is found, contact the local service centre for repairs.

4.2.2 Monthly examination of the door



Do not use the door leaf or the track set to support a ladder when you do maintenance on a door. Always use ladders as specified in local health and safety instructions.

1) Use a soft brush and a mild detergent to clean the track set and the door seals.
2) Make sure there are no loose screws, bolts or nuts on the door leaf or the track set.
3) If necessary tighten all loose screws, bolts and nuts.
4) Examine all door leaf hinges, door seals, rollers and roller holders for damage.
5) If damage is found, contact the local service centre for advice.
6) Examine the door cables for damage and corrosion.
7) If damage or corrosion is found, contact the local service centre for advice.
8) Lubricate the metal door-leaf hinges with oil (SAE 20).
9) Remove all tools and equipment from the area.
10) Do the Daily Start Procedure.

5. Tests and Adjustments

5.1 Test the balancing system

5.1.1 Electrical

1) Make sure the Daily Start Procedure has been done.



Make sure there are no persons or equipment in the working area of the door before you operate the door.

2) Make sure the pass-door is closed correctly.
3) Release the door lock.
4) Pull the rope with the green ball to disengage the operator clutch.
5) Carefully pull up on the rope or the handle until the door is open approximately 50cm.
6) The door should stay in this position.
7) If the door does not stay in the correct position:
   1) Carefully pull down on the rope and the handle to close the door.
   2) Contact the local service centre for advice.
8) Carefully pull down on the rope and the handle to close the door.
9) Pull the rope with the red ball to engage the operator clutch.

5.1.2 Pull-down rope or chain hoist operation

1) Make sure the Daily Start Procedure has been done.



Make sure there are no persons or equipment in the working area of the door before you operate the door.

2) Make sure the pass-door is closed correctly.
3) Release the door lock.
4) Carefully pull up on the rope or the handle until the door is open approximately 50cm.



Pulling on the wrong side of the chain hoist when the door is closed can cause serious damage. Always make sure that you are pulling on the right side of the chain.

5) The door should stay in this position.
6) If the door does not stay in the correct position:
   1) Carefully pull down on the rope and the handle to close the door.
   2) Contact the local service centre for advice.

6. Troubleshooting

6.1 Introduction

This chapter contains troubleshooting information for users of this door. If a fault is not described in this chapter contact your local Service Centre for assistance.

6.2 Electrically operated door

6.2.1 Door does not go up or down correctly

Possible Cause	Solution
Damage to door or track set	Do the monthly examination of the door
Balancing system not operating correctly	Test the balancing system
No main electrical supply	Switch on electrical supply
Emergency stop button pressed in	Make sure the Daily Start Procedure has been done.
Error code displayed on control unit (950 door control system only)	Refer to the error codes list and do the applicable action.
Operator clutch disengaged	Engage the operator clutch as follows: 1) Pull the rope with the green ball to disengage the operator clutch. 2) Pull the rope with the red ball to engage the operator clutch. 3) Press and release the  button and make sure the door moves to the fully open position.

6.2.2 Error codes list

On the 950 door control system error codes appear on the digital display (see page 8). The following errors can be solved by the user. For any other codes, please contact your local Service Centre for assistance.

Error Code	Cause of error	Solution
E04	Safety switch error	Examine the spiral cable and the conduit for damage. Make sure the conduit is connected to the bottom seal.
E09	Mains voltage too low	Make sure the correct mains voltage is supplied to the control unit
E11	Operator clutch disengaged	Engage operator clutch as follows: 1) Pull the rope with the green ball to disengage the operator clutch. 2) Pull the rope with the red ball to engage the operator clutch. 3) Press and release the  button and make sure the door moves to the fully open position.
E24	Door out of position	1) Do the Daily Start Procedure 2) Press and release the  button and make sure the door moves to the fully open position.

6.3 Manual or chain hoist operated door

6.3.1 Door does not go up or down correctly

Possible cause	Solution
Damage to door or track set	Do the monthly examination of the door
Balancing system not operating correctly	Test the balancing system

7. Recycling

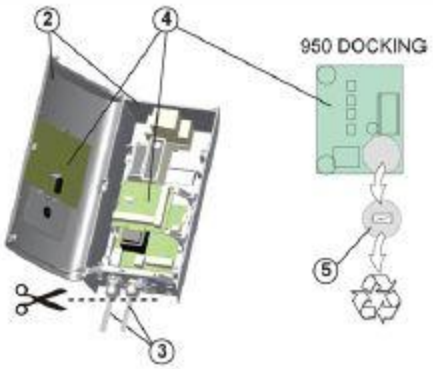
7.1 Recycling the Operator

The power has to be switched off before any work at the product.

The product should be returned for recycling at your local recycling depot. N.B. Upon disposal, the material components of the product such as metal, plastic, wiring, circuit boards and battery should be separated before it is deposited for recycling.

1) Metal
2) Plastic
3) Wiring
4) Circuit boards
5) Battery





The battery must be removed from the appliance before it is scrapped. The appliance must be disconnected from the supply mains when removing the battery. The battery has to be disposed safely.